30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 12 (Ch. 8)

Phillips Curve

MICHELA BRAGA

WHERE WE ARE

Fundamentals

1. Macroeconomic variables
   - Elements of national accounting

2. The short run
   1. Goods markets
   2. Financial markets
      1. Without commercial banks
      2. With commercial banks
   3. IS-LM model

3. The medium run
   1. Labor market
   2. Phillips curve
   3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
   1. Financial markets and expectations
   2. Consumption, investment, and expectations
   3. Output, economic policies, and expectations

5. Open economy
   1. Exchange rates, trade, and international returns
   2. Goods markets in an open economy
   3. Financial markets and economic policies in an open economy

6. Public debt and fiscal policy

7. From the Great Recession to Quantitative Tightening
   - From a housing crisis to a financial crisis
   - Unconventional monetary policies

SHORT RUN

Transition occurs thanks to changes in prices, wages and adjustment of expectations

MEDIUM RUN

- Wrong expectations $P^e \neq P$
- Goods and financial mkt in equilibrium
- Economy not at the potential $Y \neq Y_n$

There is a link btw labor market and price dynamics

- Correct expectations $P^e = P$
- Labor market in equilibrium: $u = u_n$
- Economy at its potential $Y = Y_n$

Phillips Curve
empirical relation between unemployment and inflation rate

THE LABOR MARKET

Let's consider the labor market

- in the short run: $P^e \neq P$
- assuming unitary labor productivity: $A = 1 \Rightarrow$ production function $Y = N$

The two key equations studied in the previous lecture become

1. Wage setting (WS):

$$W = P^e F(u, z)$$

2. Price setting (PS):

$$P = (1+m)\frac{W}{A} = (1+m)W$$

Substituting the WS into the PS we obtain

$$\boxed{P = (1+m)P^e F(\underset{-}{u}, \underset{+}{z})}$$

Equation describing the determinants of the current price level in the economy based on WS-PS model

- $m \uparrow \Rightarrow$ firms have higher mkt power and charge higher prices $\Rightarrow$ $P \uparrow$
- $P^e \uparrow \Rightarrow$ workers ask and obtain higher nominal wages $\Rightarrow$ if $W \uparrow \Rightarrow \uparrow$ firms' production costs $\Rightarrow P \uparrow$
- $u \uparrow \Rightarrow$ workers bargaining power $\downarrow$ wrt firms' $\Rightarrow$ $W \downarrow \Rightarrow P \downarrow$
- $z \uparrow \Rightarrow$ workers bargaining power $\uparrow \Rightarrow W \uparrow \Rightarrow P \uparrow$

Assume a specific linear functional form for $\mathrm{F}(\mathrm{u}, \mathrm{z})$

$$F(u,z) \equiv 1 - \alpha u + z$$

$\alpha > 0$

sensitivity of wages to labor market conditions

If $\alpha \uparrow$, *W* vary more when *u* changes

Substitute $F(u,z) = 1 - \alpha u + z$ in the equation for prices we obtain

$$P = (1+m)\, P^e (1 - \alpha u + z)$$

AIM: explain the dynamics from short to medium run => Add time to the equation

m, z, α exogenous parameter

$$P_t = (1+m)\ P_t^e\ (1 - \alpha\ u_t + z)$$

Actual unemployment

Equation describing the relationship btw price level and unemployment rate

A LITTLE OF MATH TO CONSTRUCT THE PHILLIPS CURVE

Divide by previous period prices: P_{t-1}

$$\frac{P_t}{P_{t-1}} = \frac{P_t^e}{P_{t-1}}(1+m)(1-\alpha u_t + z)$$

And recalling that

$$\frac{P_t}{P_{t-1}} = 1 + \pi_t \quad \& \quad \frac{P_t^e}{P_{t-1}} = 1 + \pi_t^e$$

We obtain

$$(1+\pi_t) = (1+\pi_t^e)(1+m)(1-\alpha u_t + z)$$

relationship btw inflation rate and unemployment rate

Taking logs of both sides, and assuming small values of the variables we have we can approximate the previous equation as

$$\pi_t \approx \pi_t^e + m - \alpha u_t + z$$

$$\boxed{\pi_t = \pi_t^e + (m + z) - \alpha u_t}$$

GENERAL Phillip curve

No assumptions on how agents form their expectations

Describes the determinants of inflation in the economy:

1. Price expectations (+)
2. Labor mkt conditions/unemployment (-)
3. Structural factors (+)
   - Of the labor market: z
   - Of the products mkt: m

From the general Phillips curve we know that $\pi_t = \pi_t^e + (m+z) - \alpha u_t$

- $\boldsymbol{u \downarrow \Rightarrow \pi_t \uparrow}$:
  - Better conditions on labor market
  - Workers have more bargaining power
  - Workers ask for higher wage
  - Prices increase
  - The impact is larger, the larger α

Policy makers face a trade –off:
to reduce unemployment, they have to accept higher inflation

- $\boldsymbol{\pi^e \uparrow \Rightarrow \pi_t \uparrow}$:
  - People expect higher growth in prices
  - Ask for higher nominal wages
  - Firms production costs increase
  - Prices level increases

- $\boldsymbol{m \uparrow \Rightarrow \pi_t \uparrow}$:
  - Lower competition on product markets
  - Firms have more market power and charge higher prices
  - Prices grow faster

- $\boldsymbol{z \uparrow \Rightarrow \pi_t \uparrow}$:
  - Higher workers' bargaining power
  - Workers ask for higher wage growth
  - Production costs increase faster
  - Prices grow faster

What is the unemployment rate if expectations are correct?

Let's rearrange the PC

$$\Delta\pi = \pi_t - \pi_t^e = (m+z) - \alpha u_t$$

If $\pi_t = \pi_t^e$, $\Delta\pi = 0$

$$0 = (m+z) - \alpha u_t$$
$$\alpha u_t = (m+z)$$

$$\boxed{u_t = \frac{m+z}{\alpha}}$$

In the medium run, $P^e = P \Rightarrow \pi = \pi^e$

u_n= Natural (structural) rate of unemployment

Only structural factors matter

It depends on the structure of the economy as we have seen in the WS-PS model

$$m \uparrow \Rightarrow u_n \uparrow$$
$$z \uparrow \Rightarrow u_n \uparrow$$
$$\alpha \uparrow \Rightarrow u_n \downarrow$$

Also productivity A if we do not assume it $= 1$

We can write the PC in terms of natural unemployment rate

$$\pi_t = \pi_t^e + (m+z) - \alpha u_t$$

$$\pi_t - \pi_t^e = (m+z) - \alpha u_t$$

$$\pi_t - \pi_t^e = -\alpha\left(u_t - \boxed{\left(\frac{m+z}{\alpha}\right)}\right) \rightarrow u_n$$

$$\boxed{\Delta\pi = \pi_t - \pi_t^e = -\alpha(u_t - u_n)}$$

Phillips curve as a relationship btw the gap btw current unemployment and the gap btw current and expected inflation

- If the economy is at the potential: $Y_t = Y_n$ and $u_t = u_n$

Inflation is at the expected rate $\Delta\pi = 0$

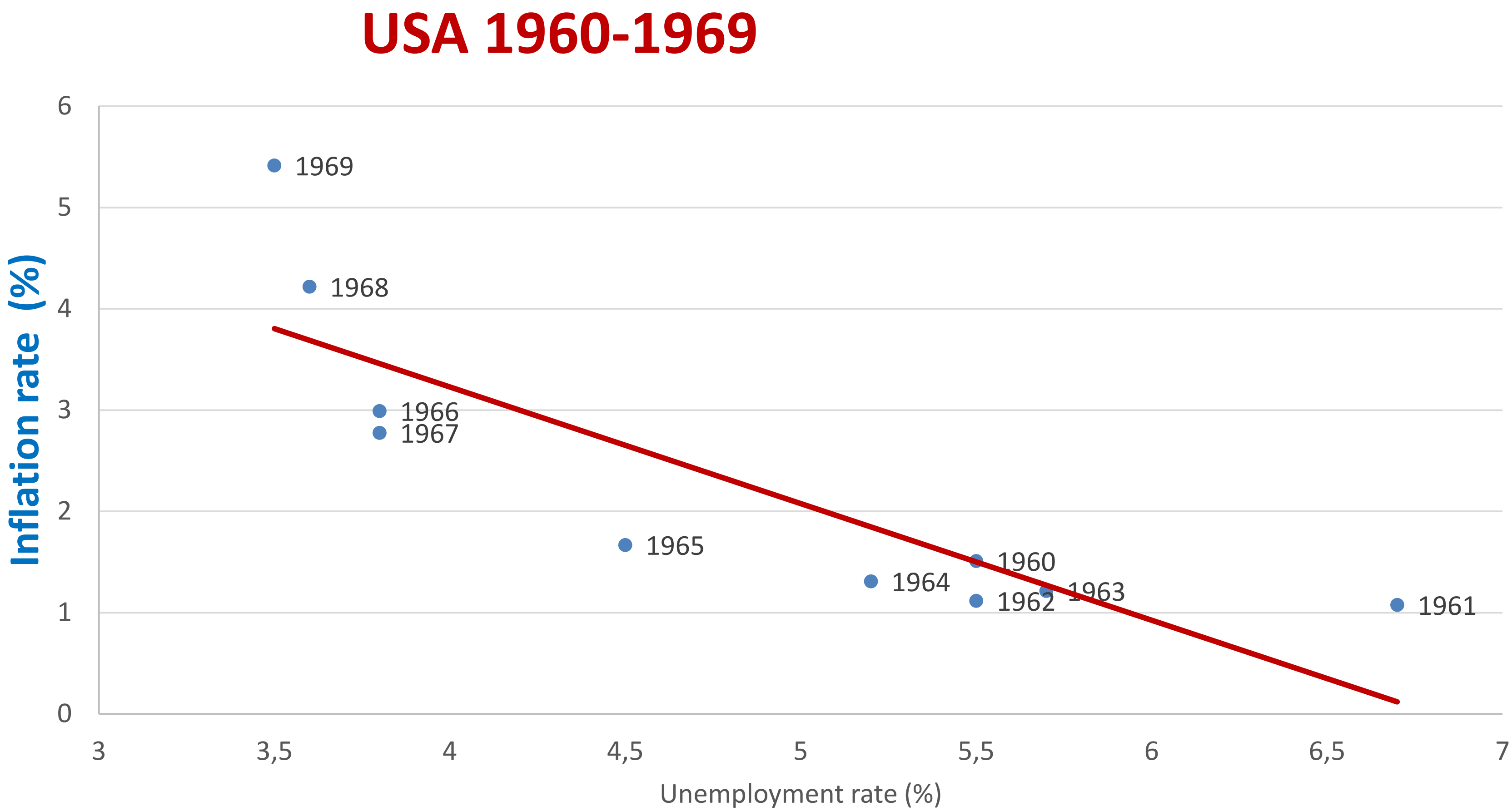
- If the economy is above the potential: $Y_t > Y_n$ and $u_t < u_n$

Lower unemployment wrt. MR (expansion) ⇒ Faster growth of W ⇒ P growth increases ⇒ inflation higher than expected $(\Delta\pi_t > 0)$

- If the economy is below the potential: $Y_t < Y_n$ and $u_t > u_n$

Higher unemployment wrt. MR (recession) ⇒ slower growth of W ⇒ P growth decreases ⇒ inflation lower than expected $(\Delta\pi_t < 0)$

PHILLIPS CURVE: empirical evidence



Data confirm the model !!!!

PHILLIPS CURVE: empirical evidence

ORIGINAL PHILLIPS CURVE: empirical evidence

- Original PC empirically supported up to 1960s
- From 70s on, no empirical support
- What happened???
  - Agents changed the way in which they form expectations
  - The «shape» of the PC (the relationship btw inflation and unemployment) depends on how agents form their expectations ⇔ how they use the available information

EXPECTATIONS:
ANCHORED VS. ADAPTIVE

Anchored expectations: we expect “something similar” today to what we have seen in the past

Adaptive expectations: we expect the same inflation today as yesterday

A POSSIBLE GENERAL RULE FOR EXPECTATION

$$\pi_t^e = (1-\theta)\ \overline{\pi}\ \ + \theta\ \ \pi_{t-1}$$

Inflation observed in the last period

Constant inflation level
- average inflation before t-1
- CB inflation target

$0 \leq \theta \leq 1$ inflation persistency over time

If $\theta\uparrow$, inflation more persistent ⇔ previous period's inflation has a greater influence on current inflation ⇔ workers place more weight on the previous period's inflation when bargaining their wages.

The PC can be written as

$$\boxed{\pi_t = (1-\theta)\ \overline{\pi}\ \ + \theta\ \ \pi_{t-1} + (m+z) - \alpha u_t}$$

1) ANCHORED EXPECTATIONS

- Assumption: agents expect an inflation rate equal to a value $\bar{\pi}$ (the CB target)

$$\boldsymbol{\pi^e = \bar{\pi}}$$

- Not persistent inflation: θ=0

- Substituting into the general formulation :

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

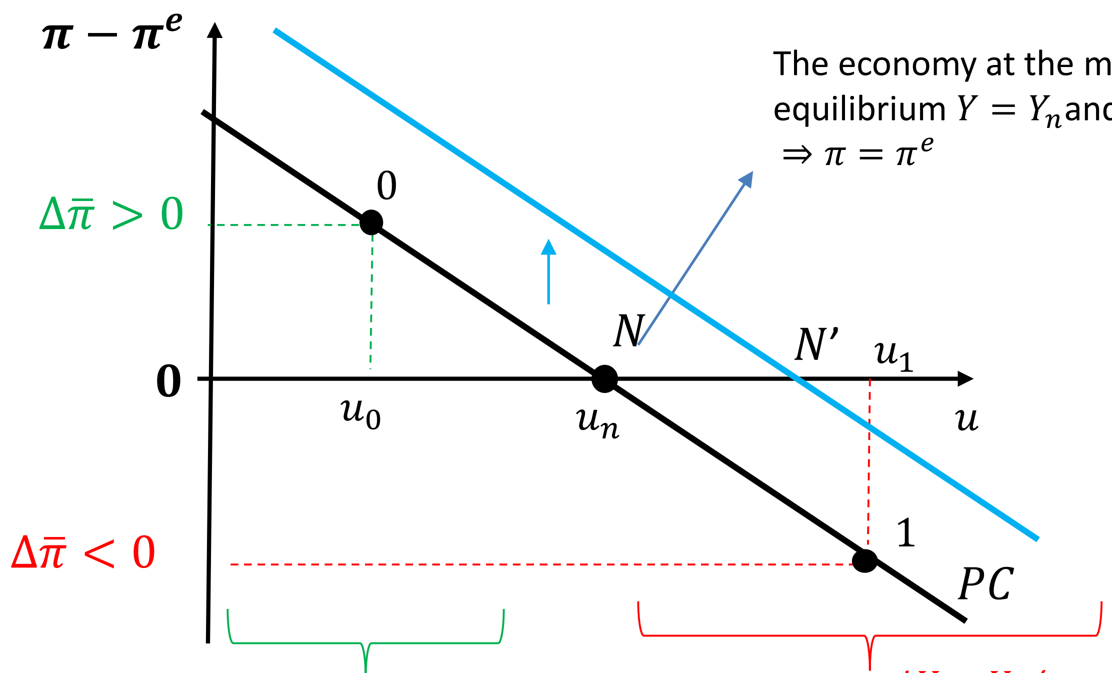
$$\Delta\bar{\pi} = \pi_t - \bar{\pi} = (m + z) - \alpha u_t$$
$$= -\alpha(u_t - u_n)$$

Original /Anchored PC

Negative relationship btw unemployment rate and LEVEL of inflation

Assume the CB has an inflation target $\bar{\pi}$ => If it is credible, we form expectations accordingly $\pi^e = \bar{\pi}$

$$\Delta\bar{\pi} = \pi_t - \bar{\pi} = (m + z) - \alpha u_t$$
$$= -\alpha(u_t - u_n)$$



$u < u_n$ and $Y > Y_n$ (positive O.G.)
$\Delta\bar{\pi} > 0 \;\Rightarrow\; \pi > \bar{\pi}$ => inflation higher than expected since workers ask for higher wages
$\Rightarrow$ short run equilibrium

$u > u_n$ and $Y < Y_n$ (negative O.G.)
$\Delta\bar{\pi} < 0 \;\Rightarrow\; \pi < \bar{\pi}$ => inflation lower than expected
$\Rightarrow$ short run equilibrium

Note: The PC shifts up if ↑z and/or ↑m => taking constant u, current inflation increases wrt the target => change in u_n and Y_n (structural shock)

ADAPTIVE EXPECTATIONS

- During 70s inflation became **persistent**
  - Inflation in a period seems to be a good predictor of inflation in the next period

- Adaptive expectations
  - determined (at least in part) by observed prices

2) ADAPTIVE EXPECTATIONS

- Assumption: agents expect an inflation rate equal to the one of the last period

$$\boldsymbol{\pi^e} = \pi_{t-1}$$

- Persistent inflation: θ=1

- Substituting into the general formulation :

$$\boxed{\begin{array}{c} \pi_t = \pi_{t-1} + (m+z) \ - \alpha u_t \\ \\ \Delta\pi = \pi_t - \pi_{t-1} = (m+z) - \alpha u_t \\ = -\alpha(u_t - u_n) \end{array}}$$

Modified/ Accelerationist PC

Negative relationship btw unemployment rate and CHANGE in inflation

If $u_t = u_n \ Y_t = Y_n \Rightarrow \ \Delta\pi_t = 0$: inflation is constant (medium run equilibrium)

If $u_t > u_n \ Y_t < Y_n \Rightarrow \Delta\pi_t < 0$: inflation decreases (short run equilibrium)

If $u_t < u_n Y_t > Y_n \Rightarrow \Delta\pi_t > 0$: inflation increases (short run equilibrium)

INTUITION

Trade-off between unemployment and **<u>change</u> of the inflation rate**

- If $u_t < u_n Y_t > Y_n$
  - Workers ask for higher wage growth
  - Prices grow faster
  - Agents expect higher inflation
  - Ask for higher wage growth
  - Prices grow faster

acceleration of inflation

WAGE INDEXATION

Mechanism of automatic adjustment of wages to inflation

AIM?

Maintain workers' purchasing power when inflation becomes high and volatile

=> Nominal wages lose purchasing power

=> more difficult to predict and agents reluctant to sign long-term contract

Impact?

- Larger impact of unemployment on inflation
- Inflation increases even faster

ASSUMPTIONS

2 types of contracts in the economy:

Not indexed

- wage set according to expectations

$$\pi_t^e = \pi_{t-1}$$

- A fraction $(1-\lambda)$ of contracts

Indexed

- wages adjust immediately to current prices

$$\pi_t^e = \pi_t$$

- A fraction λ of contracts

Expected inflation in the economy will be:

$$\pi_t^e = (1-\lambda)\pi_{t-1} + \lambda\pi_t$$

PC WITH INDEXED WAGES

- Substituting π_t^e into the general Phillips Curve:

$$\pi_t = (1-\lambda)\pi_{t-1} + \lambda\pi_t - \alpha(u_t - u_n)$$

$$(1-\lambda)(\pi_t - \pi_{t-1}) = -\alpha(u_t - u_n)$$

$$(1-\lambda)\Delta\pi_t = -\alpha(u_t - u_n)$$

$$\boxed{\Delta\pi_t = -\frac{\alpha}{1-\lambda}(u_t - u_n)}$$

Indexed PC

WAGE-PRICE SPIRAL

- An increase in prices in t has an immediate impact on wages in t

- $\frac{1}{1-\lambda} > 1$: large impact of unemployment on inflation than in the basic case

INTUITION

- Indexation amplifies the effect of u on π :

An increase in prices in t has an immediate effect on wages and therefore a further effect on prices => inflation will be higher

- Why?

Suppose that $u \downarrow$ so that the workers bargaining power of workers $\uparrow$. Firms production costs $\uparrow$ and to maximize profits they charge higher prices. The current price level increases.

  - Workers with no indexation do not experience further changes in W.
  - Workers with indexation experience an adjustment of their contracts. Firms production costs $\uparrow$ and also prices $\uparrow$. The current price level increases.

With indexation, inflation varies more as a result of the wage bargaining and of the automatic adjustment of wages to prices

OBSERVATIONS

- If the share or indexed wages in the labor force increases, inflation will be higher
- If «many» contracts are indexed => a stronger wage – inflation spiral occurs
  - «a large portion» of nominal wages follows price changes, making inflation more volatile:
  - If u ↓, W↑ => ↑ production costs for firms => ↑P => ↑ W => ↑P

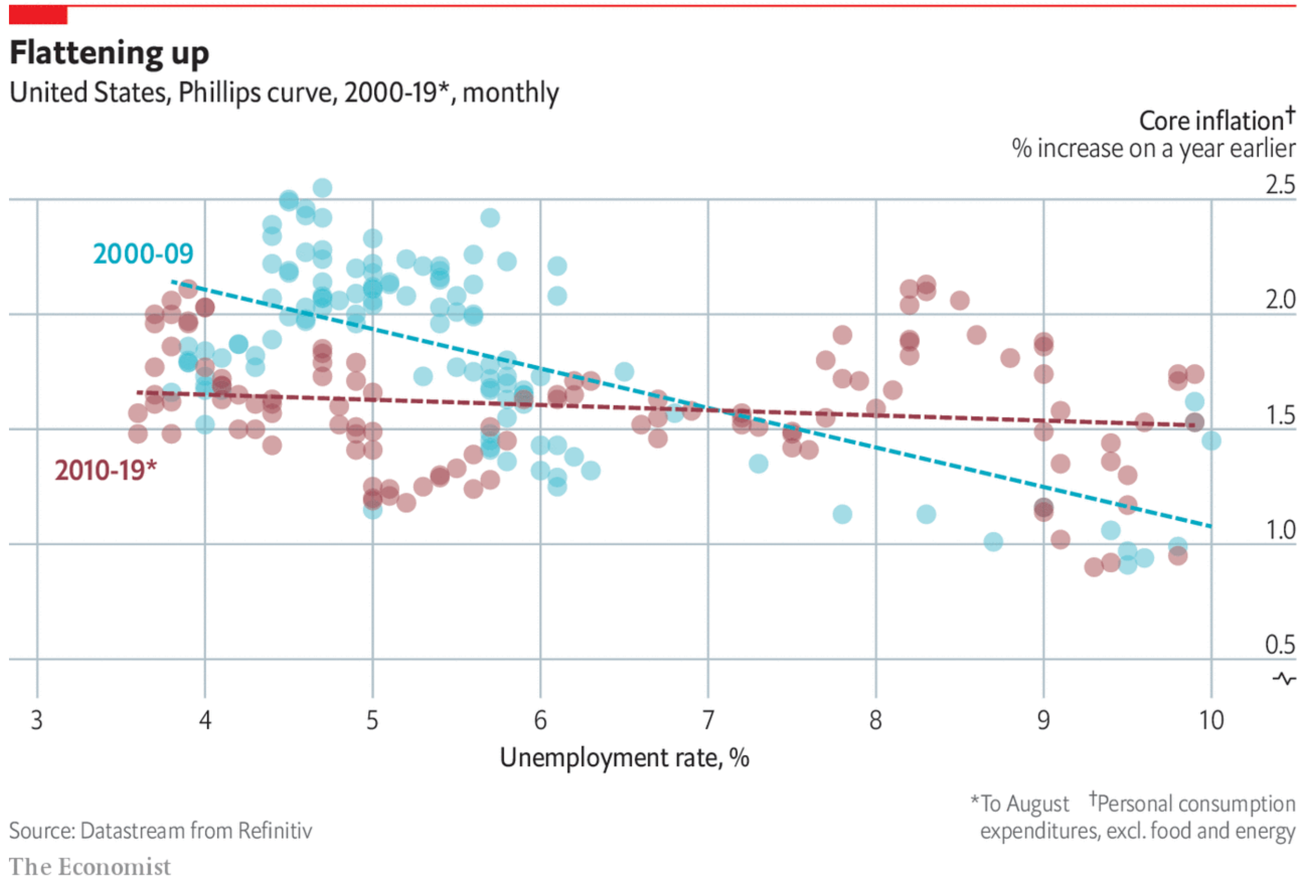
Effect of the bargaining process on all the workers

Effect of indexation on “many” workers

=> ↑ ↑ P

- If «almost all contracts» are indexed, also very «small» variations in unemployment lead to «large» changes in inflation

FLATTERING UP PHILLIPS CURVE



π less sensitive to changes in u

PHILLIPS CURVE AND DEFLATION

When $\pi < 0$ (deflation) or very low inflation

the Phillips Curve is no longer found in data

$\Leftrightarrow$

Even if $u_t > u_n$, prices do not decrease according to our prediction

- *Why?*
    - It is relatively easy to reduce the growth rate of nominal wages
    - It is more difficult to reduce nominal wages

CONCLUSION

- Phillips Curve and natural rate of unemployment

$$\pi - \pi^e = -\alpha(u_t - u_n)$$

- The PC can change over time due to the way in which agents form expectations

- Structural changes in the labor and product market affect u_n and the PC

- Countries may be characterized by a different PC (different structure of economies)

QUICK CHECK

- The accelerationist Phillips curve is a relation between …

a) change in inflation and nominal GDP.

b) change in inflation and natural interest rate.

c) effective unemployment rate and natural unemployment rate.

d) change in inflation and unemployment rate.

QUICK CHECK

- The natural rate of unemployment

a) is equal to 0 when the economy is expanding

b) is high when markets are highly competitive

c) is low when firms reduce their markups

d) is “natural” and therefore independent of the institutional characteristics of the labor market.

QUICK CHECK

A reform that reduces minimum wage, in the short run...

a) increases inflation, decreases actual unemployment, and increases natural output.

b) increases inflation, does not affect actual unemployment, and increases natural output.

c) decreases inflation, decreases actual unemployment, and increases natural output.

d) decreases inflation, does not affect actual unemployment, and increases natural output.

QUICK CHECK

- If expectations are fully adaptive, an increase in actual employment would lead to:

a) an acceleration in inflation.

b) a deceleration in inflation.

c) hyperinflation.

d) deflation

TAKE AWAY AND KEY CONCEPTS

- General PC in terms of
  - unemployment
  - gap btw actual and natural unemployment

- General rule for expectations
  - Persistence of inflation

- Expectations
  - Anchored
  - Adaptive

- PC with indexed wages
- Central Bank inflation target

FOR THOSE WHO WANT DIVE DEEPER… BUT NOT COMPULSORY

CONSTRUCTING THE PHILLIPS CURVE

$$(1+\pi_t) = (1+\pi_t^e)(1+m)(1-\alpha u_t + z)$$

…Taking logs of both sides

$$log(1+\pi_t) = log(1+\pi_t^e) + log(1+m) + log(1-\alpha u_t + z)$$

…and for small values, we can approximate as

$$\pi_t \approx \pi_t^e + m - \alpha u_t + z$$

No assumptions on how agents form their expectations

$$\boxed{\pi_t = \pi_t^e + (m+z) - \alpha u_t}$$

GENERAL Phillip curve

Describes the determinants of inflation in the economy:

- Price expectations (+)
- Labor mkt conditions/unemployment (-)
- Structural factors (+)

Phillips Curve and expectations

Original PC ($\boldsymbol{\pi^e = \bar{\pi} = 2\%}$)

Anchored expectations

- $\pi_t = \bar{\pi} + (m + z) - \alpha u_t$

  $\pi_1 = 2 + 6 - 6 = 2\%$
- Assume we want to reduce u: $u_t = 4\%$ from $t \geq 2$
  - $\pi_2 = 2 + 6 - 4 = 4\%$
  - $\pi_3 = 2 + 6 - 4 = 4\%$
  - $\pi_4 = 2 + 6 - 4 = 4\%$
  - $\pi_5 = 2 + 6 - 4 = 4\%$
- Inflation is constantly above the target
- Trade-off btw. u and inflation rate (π)

Modified PC ($\boldsymbol{\pi^e = \pi_{t-1} = 2\%}$)

Adaptive expectations

- $\pi_t = \pi_{t-1} + (m + z) - \alpha u_t$

  $\pi_1 = 2 + 6 - 6 = 2\%$
- Assume we want to reduce u: $u_t = 4\%$ from $t \geq 2$
  - $\pi_2 = \mathbf{2} + 6 - 4 = 4\%$
  - $\pi_3 = \mathbf{4} + 6 - 4 = 6\%$
  - $\pi_4 = \mathbf{6} + 6 - 4 = 8\%$
  - $\pi_5 = \mathbf{8} + 6 - 4 = 10\%$
- Inflation increases over time
- Trade-off btw. u and **change** of the inflation rate ($\Delta\pi$)